FINITENESS AND INVARIANCE OF MOD P COHOMOLOGY OF A PROPER SCHEME IN CHARACTERISTIC P

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both parts of the statement.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Cohomology with coefficients Z/p of a proper scheme over an algebraically closed field. Same chapter, the basic properties of etale cohomology with finite constant coefficients on a proper scheme. Supplies the groups whose finiteness and invariance are asserted.
- Artin-Schreier sequence and the structure of the cohomology of the structure sheaf. Same chapter, the Artin-Schreier sequence in characteristic p and the resulting description of the cohomology with coefficients Z/p in terms of differential forms and the Frobenius action. Supplies the computations which give finiteness and invariance under extensions of the base field.

Source reference. Original lemma 02O6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9878–9878. Navigation pointer invoked at proof line 9878; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 03P2: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2280–2280. Navigation pointer invoked at proof line 9880; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0A3L: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9641–9641. Navigation pointer invoked at proof line 9858; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0HAN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9692–9692. Navigation pointer invoked at proof line 9894; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02KH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9898–9898. Navigation pointer invoked at proof line 9898; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be a proper scheme over an algebraically closed field k of characteristic $p > 0$. Then*

(1) *$H^q_{étale}(X, \underline{\mathbf{Z}/p\mathbf{Z}})$ is a finite $\mathbf{Z}/p\mathbf{Z}$-module for all q, and*
(2) *$H^q_{étale}(X, \underline{\mathbf{Z}/p\mathbf{Z}}) \to H^q_{étale}(X_{k'}, \underline{\mathbf{Z}/p\mathbf{Z}})$ is an isomorphism if k'/k is an extension of algebraically closed fields.*

Proof. By Cohomology of Schemes, Lemma 02O6 and the comparison of cohomology of Theorem 03P2 the cohomology groups $H^q_{étale}(X, \mathbf{G}_a) = H^q(X, \mathcal{O}_X)$ are finite dimensional k-vector spaces. Hence by Lemma 0A3L the long exact cohomology sequence associated to the Artin-Schreier sequence, splits into short exact sequences

$$0 \to H^q_{étale}(X, \underline{\mathbf{Z}/p\mathbf{Z}}) \to H^q(X, \mathcal{O}_X) \xrightarrow{F-1} H^q(X, \mathcal{O}_X) \to 0$$

and moreover the $\mathbf{F}_p$-dimension of the cohomology groups $H^q_{étale}(X, \underline{\mathbf{Z}/p\mathbf{Z}})$ is less than or equal to the k-dimension of the vector space $H^q(X, \mathcal{O}_X)$. This proves the first statement. The second statement follows from Lemma 0HAN as $H^q(X, \mathcal{O}_X) \otimes_k k' \to H^q(X_{k'}, \mathcal{O}_{X_{k'}})$ is an isomorphism by flat base change (Cohomology of Schemes, Lemma 02KH). □